

Anisotropic Energetic Mesh Smoothing with a Metric Tensor Field in Norma

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Abstract

Energetic mesh smoothing (EMS) in Norma minimizes a hyperelastic energy of the mesh, with an ideal reference element per tetrahedron set by a scalar size. This note records the extension of that ideal element to a metric tensor, which prescribes three principal sizes and their directions, and the change in the energy that makes the orientation of the elements part of the objective: the energy is evaluated on the deformation gradient measured in the metric, and the stress and tangent are pulled back to the reference configuration. The formulation, its properties, the treatment of a metric that varies in space, the implementation, the regression tests, and two examples are described.

1 Formulation

The isotropic smoothing energy. For each tetrahedron the ideal reference element $\Omega_{\mathbf{X}}$ is a regular tetrahedron of edge length h , the current element $\Omega_{\mathbf{x}}$ is the element of the mesh, and $\mathbf{F} = \partial \mathbf{x} / \partial \mathbf{X}$ is the deformation gradient of the map between them. The smoothing energy of the mesh is $\Phi = \sum_e V_{\mathbf{X}}^e W(\mathbf{F}_e)$, with W an isotropic hyperelastic energy density (the Seth–Hill family in Norma) and $V_{\mathbf{X}}^e$ the volume of the ideal element. The nodal positions are the unknowns and the ideal elements are fixed during a solve. Because W is isotropic, it depends on $\mathbf{F}$ only through the invariants of $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $W(\mathbf{Q} \mathbf{C} \mathbf{Q}^T) = W(\mathbf{C})$ for every rotation $\mathbf{Q}$: rotating the ideal element changes nothing. This is harmless for a regular ideal element, which has no orientation, and fatal for an anisotropic one: an elongated element has zero energy in every orientation, so the energy cannot ask the elongation to point in a prescribed direction.

The metric tensor. An anisotropic target is a symmetric positive definite metric tensor

$$\mathbf{M} = \mathbf{R} \operatorname{diag}\left(\frac{1}{h_1^2}, \frac{1}{h_2^2}, \frac{1}{h_3^2}\right) \mathbf{R}^T, \quad (1)$$

with principal sizes h_i and a rotation $\mathbf{R} = \exp(\hat{\mathbf{v}})$, given by a rotation vector $\mathbf{v}$, whose columns are the principal directions. The length of a vector $\mathbf{e}$ measured in the metric is $\sqrt{\mathbf{e} \cdot \mathbf{M} \mathbf{e}}$, so an edge of length h_i along the i -th principal direction has unit length in the metric. Any factor $\mathbf{F}_M$ with $\mathbf{F}_M^T \mathbf{F}_M = \mathbf{M}$ maps physical vectors to metric-space vectors; Norma uses

$$\mathbf{F}_M = \operatorname{diag}(1/h_1, 1/h_2, 1/h_3) \mathbf{R}^T, \quad \mathbf{F}_M^{-1} = \mathbf{R} \operatorname{diag}(h_1, h_2, h_3). \quad (2)$$

The ideal element is the unit regular tetrahedron $\Omega_{\mathbf{Y}}$ mapped by $\mathbf{F}_M^{-1}$, that is, scaled by h_i along the global axes and then rotated by $\mathbf{R}$: $\mathbf{X} = \mathbf{F}_M^{-1} \mathbf{Y}$. Its volume is $h_1 h_2 h_3 V_{\text{unit}}$ with $V_{\text{unit}} = 1/(6\sqrt{2})$.

The energy in the metric space. The current element is also mapped into the metric space, $\mathbf{y} = \mathbf{F}_M \mathbf{x}$, and the energy is evaluated on the deformation gradient between the unit ideal element and the metric image of the current element,

$$\mathbf{F}_U = \frac{\partial \mathbf{y}}{\partial \mathbf{Y}} = \mathbf{F}_M \mathbf{F} \mathbf{F}_M^{-1}, \quad \Phi = \sum_e V_{\mathbf{X}}^e W(\mathbf{F}_{U,e}). \quad (3)$$

An element has zero energy if and only if its metric image is a unit regular tetrahedron, that is, if it is a unit element for $\mathbf{M}$ in the sense of metric-based meshing. If the current element is the ideal element rotated by $\mathbf{Q}$, then $\mathbf{F} = \mathbf{Q}$ and $\mathbf{C}_U = \mathbf{F}_M^{-T} \mathbf{Q}^T \mathbf{M} \mathbf{Q} \mathbf{F}_M^{-1}$, which equals the identity only when $\mathbf{Q}^T \mathbf{M} \mathbf{Q} = \mathbf{M}$: the orientation is penalized, except for rotations that leave the metric unchanged, as it should be. With equal sizes $\mathbf{F}_M = \mathbf{I}/h$, $\mathbf{F}_U = \mathbf{F}$, and the isotropic energy is recovered exactly.

Stress and tangent. With the ideal volume as the weight, $\Phi_e = V_{\mathbf{X}}^e W(\mathbf{F}_M \mathbf{F} \mathbf{F}_M^{-1})$, and the chain rule gives the first Piola–Kirchhoff stress and the tangent in the reference configuration from those of the isotropic model evaluated at $\mathbf{F}_U$,

$$\mathbf{P} = \mathbf{F}_M^T \mathbf{P}_U \mathbf{F}_M^{-T}, \quad A_{iJkL} = (F_M)_{pi} (F_M^{-1})_{Jq} A_{pqrs}^U (F_M)_{rk} (F_M^{-1})_{Ls}, \quad (4)$$

where $\mathbf{P}_U = \partial W / \partial \mathbf{F}_U$ and $\mathbf{A}^U = \partial \mathbf{P}_U / \partial \mathbf{F}_U$. Nothing else in the assembly changes. The same result follows from the nodal chain rule: with $\mathbf{y}_a = \mathbf{F}_M \mathbf{x}_a$, the nodal forces are $\mathbf{f}_x = \mathbf{F}_M^T \mathbf{f}_y$ and the nodal stiffness blocks $\mathbf{K}_x = \mathbf{F}_M^T \mathbf{K}_y \mathbf{F}_M$.

Properties.

1. The energy does not depend on the factor chosen for $\mathbf{M}$: any other factor is $\mathbf{Q} \mathbf{F}_M$ with $\mathbf{Q}$ a rotation, which enters $\mathbf{F}_U$ as a superimposed rigid rotation and leaves the isotropic W unchanged. The eigenvector signs and the in-plane phase of $\mathbf{R}$ for repeated sizes are therefore irrelevant to the energy.
2. The orientation of the unit tetrahedron $\Omega_{\mathbf{Y}}$ is irrelevant for the same reason: every rotation of the unit tetrahedron mapped by $\mathbf{F}_M^{-1}$ is a zero-energy element. The smoother does not prefer one packing of anisotropic elements over another; the mesh topology decides.
3. $\det \mathbf{F}_U = \det \mathbf{F}$, so the volumetric part of the energy is unchanged by the metric transformation: it compares the element volume with $h_1 h_2 h_3 V_{\text{unit}}$.

2 A metric that varies in space

The metric is evaluated once per element at the centroid of the element in the original mesh, as the isotropic size field is, and held fixed during a solve. The energy is then an exact function of the nodal positions and the assembled force is its exact gradient, which the line search and the convergence tests of the solver require. If the metric followed the current centroid, $\mathbf{F}_M$ would depend on the unknowns and the exact gradient would contain the derivative of $\mathbf{F}_M$, which involves the derivative of the exponential map of the rotation vector; omitting it would make the force inconsistent with the energy. The metric is therefore Lagrangian: an element keeps the target it was assigned where it started. For smoothing, where nodes move a fraction of an element, this is a second-order effect. For large motions, or when smoothing alternates with topological operations, the targets are

refreshed at phase boundaries from the field at the current positions, which is the re-baselining of the adaptivity note (`docs/notes/ems-adaptivity`); each phase then minimizes a fixed functional.

The metric takes the pseudo-time t as an argument, so a target can be continued from the sizes of the original mesh to the desired ones over the pseudo-time steps of the quasi-static solve. Because the sizes and the rotation vector are given separately, a linear path in $\log h_i$ and in $\mathbf{v}$ interpolates the sizes geometrically and stays a rotation, which is the interpolation the metric literature recommends.

The volume floor. Smoothing cannot create elements. The isotropic `size field` rule therefore takes the larger of the size field and the edge length of the regular tetrahedron with the volume of the original element, so that no element is asked to shrink below what the mesh topology can accommodate; without it the size field produced slivers. The rule `metric field` applies the same floor to the metric: when $h_1 h_2 h_3 V_{\text{unit}}$ is smaller than the volume of the original element, the three sizes are scaled by the same factor until the volumes match, so the shape and the orientation of the target are preserved. The rule `metric field unrestricted` omits the floor.

3 Metrics given as tensors and carried by the nodes

The sizes and the rotation vector are the natural input when the target is designed by hand. A metric that comes from an error estimator, from a previous run, or from another code comes as a tensor field, usually at the nodes of the mesh. Norma accepts the metric in four forms: the sizes with an optional rotation vector, the six components of the tensor, as expressions in t, x, y, z ; and the same two forms as nodal variables of the input mesh, read at a chosen time step.

Factoring a tensor. The energy needs $\mathbf{F}_M$ with $\mathbf{F}_M^T \mathbf{F}_M = \mathbf{M}$; by property 1 of Section 1 any factor gives the same energy, so a tensor given by its components is factored by the eigendecomposition $\mathbf{M} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T$, $h_i = \lambda_i^{-1/2}$, $\mathbf{R} = \mathbf{V}$ with the sign of one column fixed by $\det \mathbf{R} = 1$, and no convention on the frame enters the input. A tensor that is not positive definite is rejected.

Interpolation between nodes. A metric carried by the nodes must be interpolated to the element. The element value is the mean over the four nodes: of the logarithms of the sizes, which interpolates the sizes geometrically, and of the rotation vectors, which stays a rotation. This form keeps the sizes exact where the frame turns: two nodes with the same sizes and different directions give an element with the same sizes and the mean direction. The alternative, the mean of the logarithms of the tensors (log-Euclidean interpolation), preserves the determinant but not the sizes: two tensors with the same sizes and frames turned by 90° average to a rounder tensor. It needs no frame, however, which is why it remains as the fallback below.

Following the frame. The rotation vectors can be averaged only when the frames of adjacent nodes are the same up to a small rotation. A tensor determines its frame only up to the 24 proper signed permutations of its eigenvectors (fewer when sizes are repeated), and an eigensolver picks one of them arbitrarily at every node. The sizes and the rotation vector are therefore recovered from nodal tensors with a frame that follows the mesh: the nodes are visited breadth first through the element adjacency, and at each node the frame closest to the frame of the node it was reached from is chosen (the largest trace of $\mathbf{R}_{\text{parent}}^T \mathbf{R}$ over the equivalents; the first node of a component takes the frame closest to the global axes). Repeated sizes leave the eigenvectors of the repeated pair free in their plane; the in-plane phase is taken from the projection of the parent frame onto that

plane, and an isotropic tensor takes the parent frame. The rotation vector is then continued past a half turn from the parent's, so a frame that turns by more than π along a path keeps a continuous vector. Where the frame turns by more than 45° between adjacent nodes after this choice, the turn cannot be absorbed by a change of representation: the field has a genuine discontinuity or a singular point of its frame. The turns are counted and reported, and the input `interpolation: log-Euclidean` switches to the interpolation of the tensor logarithms, which is insensitive to the frame.

Through the adaptivity loop. A metric carried by the nodes is Lagrangian by construction: the node keeps its target when it moves. Under the topological operations of the adaptivity note, a node added by an edge split takes the mean of the two ends, in the space the source interpolates in (logarithms of sizes and rotation vectors, or logarithms of tensors), a collapse drops the removed node, and the compaction of the mesh compacts the nodal data with it. The adapted mesh is written with the nodal variables under the names the input gave them, so the input file applies unchanged to the adapted mesh. The function sources are simply re-evaluated where the new elements lie. The size operations of the loop (collapse below $1/\sqrt{2}$, split above $\sqrt{2}$ in the metric) measure an edge with the metric of its midpoint for the function sources and with the mean of its two ends for the nodal sources.

4 Implementation

- `MetricField` (`src/model_types.jl`): a source and the `restricted` flag; stored on the model as `metric_field`. The sources are `PrincipalMetricFunctions` (compiled size and rotation-vector expressions in t, x, y, z), `TensorMetricFunctions` (six compiled components), `NodalPrincipalMetric` (logarithms of the sizes and rotation vectors at the nodes, with the names of the variables they came from), and `NodalTensorMetric` (logarithms of the tensors at the nodes).
- `create_metric_field` (`src/metric.jl`): reads the `metric field` block under `model`: exactly one of `sizes` (with optional `rotation vector`), `tensor`, `nodal sizes` (with optional `nodal rotation vector`), or `nodal tensor`, the last two from the input mesh at `time index` (default: the last step) with `interpolation: principal` (default, frames followed by `principal_of_nodal_tensors`) or `log-Euclidean`.
- `principal_of_tensor`: the factorization with the frame closest to a reference (24 signed permutations for distinct sizes, the in-plane phase from the reference for a repeated pair, the reference itself for an isotropic tensor).
- `principal_metric`: the sizes and rotation of a source on an element, at the sampling point for the function sources and as the mean over the element's nodes for the nodal sources; `create_metric_reference` calls it, checks the sizes, applies the floor, and returns the ideal element $\mathbf{F}_M^{-1}\mathbf{Y}$ together with $\mathbf{F}_M$ and $\mathbf{F}_M^{-1}$.
- `evaluate_element!` (`src/model.jl`): with a metric field, the ideal element is built by the function above, the constitutive model is evaluated at $\mathbf{F}_U$, and the stress and the tangent are pulled back with (4) by `pull_back_tangent`. Without a metric field the code path is unchanged.
- `Adaptivity` (`src/adapt.jl`): `metric_node_data` gives a split node the mean of the ends, `metric_with_node` appends it to the source without a copy for the evaluation of the proposal,

`add_metric_node!` and `compact_metric!` keep the nodal data aligned with the topology, and `metric_nodal_variables` writes it to the adapted mesh.

- Input validation knows the keys of the block; the reference documentation is in `docs/src/reference/model.md`. The sizes and rotation vectors at the nodes are written by every source as the nodal variables `size_1..3` and `rotation_1..3` (`nodal_metric_output`, with frames followed for the tensor sources), so the output of a run can be the input mesh of a nodal run, and with them the two tensor forms (`nodal_metric_tensors`): the metric tensor $\mathbf{M} = \mathbf{R} \text{diag}(h_i^{-2}) \mathbf{R}^T$ as `metric_xx` and its five companions, and the target tensor $\mathbf{T} = \mathbf{R} \text{diag}(h_i) \mathbf{R}^T$, whose eigenvalues are the target edge lengths, as `target_xx` and its companions, which is the form a tool that reads scaled eigenvector tensors expects.

5 Tests

`test/smoothing-metric.jl` (test 126) checks:

1. the ideal element and the factor: $\mathbf{F}_M \mathbf{F}_M^{-1} = \mathbf{I}$, $\mathbf{F}_M^T \mathbf{F}_M = \mathbf{M}$ with and without rotation, unit edges of $\mathbf{F}_M \mathbf{X}$, the ideal volume, agreement with the isotropic size field for equal sizes, the floor (volume of the original element recovered, size ratios preserved), evaluation at the reference centroid and at the supplied time, and the input errors;
2. the orientation sensitivity on one element with sizes (0.25, 1, 1) rotated by 0.4 rad about z : the ideal element has zero energy, the same shape rotated by a quarter turn about z has energy above one while the isotropic energy on $\mathbf{F}$ is zero for both, a rotation about the axis of transverse isotropy of the metric keeps the energy at zero, and $\mathbf{Q} \mathbf{F}_M$ gives the same energy as $\mathbf{F}_M$;
3. the assembled force and stiffness on the cube mesh (`examples/ems/cube`, 7677 elements) under a metric with all six expressions varying in space: central finite differences of the energy match the force to 10^{-5} relative and a finite-difference directional derivative of the force matches the stiffness to 10^{-5} relative;
4. a smoothing run on the cube with a constant metric at 45° in the x - y plane (sizes $0.5H$, $1.5H$, $1.5H$ with H the mean edge length): the energy decreases, the mean deviation of the edge lengths measured in the metric from unity decreases, and no element inverts.

`test/smoothing-metric-nodal.jl` (test 132) checks:

1. the factorization: sizes and rotation recovered from $\mathbf{R} \text{diag}(h_i^{-2}) \mathbf{R}^T$ for distinct sizes, a large rotation followed when the reference is near it and replaced by a closer equivalent when the reference is the identity, the repeated pair with its in-plane phase from the reference, the isotropic case, twenty random tensors and references reproduced by proper rotations, and the rejection of an indefinite tensor;
2. the frames followed along a chain of twelve nodes whose frame turns by 0.3 rad per node, past a half turn in total, with continuous rotation vectors and no reported turn; a generic turn beyond 45° from every equivalent, found by search, reported for the two adjacent pairs it creates, with every tensor still reproduced;
3. the equivalence of the sources on the cube: the tensor given by its components, constant nodal sizes and rotation vectors, constant nodal tensors under both interpolations, all give the energy and force of the sizes and rotation vector to 10^{-10} and 10^{-8} relative; nodal sizes

without a rotation give the sizes along the axes; the variables written for an adapted mesh reproduce the input; a graded field sampled at the nodes agrees with the analytic field to 2% and not to 10^{-8} ; a missing variable, a bad time index, and a nonpositive size are rejected;

4. one adaptivity loop of two iterations on the cube with graded nodal sizes finer than the mesh, splits only: the adapted meshes carry the variables under the input names, the carried data matches the final mesh node for node, and every carried size lies within the range of the input.

The existing smoothing tests (51, 119), the adaptivity tests (127 to 131), and the input-validation sweep of the examples (122) pass unchanged.

6 Examples

Cube, constant metric at 45° (`examples/ems/cube/cube-metric.yaml`). Sizes 0.05, 0.1414, 0.1414 (the volume of the mesh size 0.1), first principal direction at 45° in the x - y plane, unrestricted rule, nodes fixed in the normal direction on the faces, L-BFGS. Over the edges of the mesh the mean of $|\|\mathbf{F}_M \mathbf{e}\| - 1|$ goes from 0.44 to 0.31, the mean edge component along the small direction from 0.055 to 0.042 (target 0.05), and the mean component across it from 0.085 to 0.098. The number of elements is fixed, so the metric cannot be met everywhere; the elements rotate and stretch toward it.

Tube, graded axial size (`examples/ems/tube/tube-metric.yaml`). The rotation vector $(0, -\pi/2, 0)$ carries the first principal direction onto the axis of the tube; the axial size grows linearly from 0.06 at the bottom to 0.18 at the top while the radial and circumferential sizes stay at the mesh size 0.12, restricted rule, nodes sliding on the four analytic surfaces. The mean axial size equals the mesh size, so the grading can be realized by sliding without a change of topology. The mean of $|\|\mathbf{F}_M \mathbf{e}\| - 1|$ over the edges goes from 0.18 to 0.12, and the mean length of the axial edges (edges with more than 70% of their length along z) in the four quarters of the height, from bottom to top, goes from 0.105, 0.109, 0.109, 0.105 to 0.075, 0.110, 0.127, 0.144, against targets of 0.075, 0.105, 0.135, 0.165 at the quarter centers. The largest nodal motion is 0.28, about two element sizes, with no element inversion.

The same metric as a tensor and from the nodes (`examples/ems/cube/cube-tensor.yaml`, `examples/ems/cube/cube-metric-nodal.yaml`). The cube metric above as the tensor $M_{xx} = M_{yy} = 225$, $M_{xy} = 175$, $M_{zz} = 50$, and as the nodal variables `size.1..3`, `rotation.1..3` of the output of `cube-metric.yaml` read at its first step. A constant field has the same mean at the nodes as its value at the centroid, so the nodal run starts from the energy and residual of the analytic run (4.32×10^4 and 8.54×10^4) and ends at the same energy (9.36×10^3). The tensor run ends at 9.37×10^3 : its component $M_{zz} = 50$ corresponds to the size 0.14142 rather than 0.1414, a difference of 0.015% in the target that the energy resolves.

What smoothing cannot do. A target that changes the number of elements across the wall or around the circumference, such as a radial size of a quarter of the wall thickness on a mesh with one element through the wall, is not reachable by smoothing: the mesh barely moves and the misfit does not decrease. Such targets require the topological operations of the adaptivity note.

The same obstruction has a sharper form on a closed domain, found while answering a question from Jay Foulk. On the unit cube meshed at 0.1, with the six faces held as planes, an axis-aligned target of sizes (0.025, 0.2, 0.2) leaves the mesh exactly isotropic: the mean edge component stays

$(0.052, 0.052, 0.052)$, and the solve stops at a strict local minimum, with the gradient below 10^{-8} and the Hessian positive definite (smallest eigenvalue 399 over 4140 free degrees of freedom). A randomly displaced initial mesh, whose worst scaled Jacobian is 0.097 and whose energy starts at 1.10×10^5 instead of 8.06×10^4 , converges to the same energy, 7.2675×10^4 : the configuration is fixed by the constraints, not by the path. The reason is arithmetic. The faces are fixed planes, so the extent across x is one, and the number of elements along a path across the mesh is what the connectivity gives, so the mean advance per element is pinned near 0.05; a target of 0.025 asks for twice as many layers as exist. Milder axis-aligned targets, $(0.05, 0.1414, 0.1414)$ and $(0.05, 0.2, 0.1)$, leave the mesh isotropic for the same reason.

The same metric rotated by 45° is partly reachable, because the thin direction is no longer normal to a pair of fixed faces and a path through the mesh can zigzag in the plane normal to it: the mean edge component along the thin direction goes from 0.052 to 0.029 against the 0.025 asked, and the energy falls by a factor of five. Ramping the sizes from the mesh size to the target over the pseudo-time of the solve reaches the same minimum as the direct target, so this is not a matter of following the minimum. The remedy is topology: the same axis-aligned target under the adaptivity loop takes the energy from 7.27×10^4 to 8.7×10^3 in two iterations, at 11835 elements from 7677.

7 Next steps

The nodal metrics, the tensor input with the frame followed through the mesh and the log-Euclidean fallback, and the coupling with the topological operations of the adaptivity note are done (Section 3). What remains:

1. Construction of the metric from a solution: the recovered Hessian of a field of interest, scaled to a prescribed error or element count, written as nodal tensors that the nodal source reads.
2. A frame that follows the mesh cannot be continuous around a singular point of the field (a swirl); there the log-Euclidean fallback applies to the whole mesh. A local switch, or a frame taken from a smoothed rotation field, would keep the exact sizes elsewhere.